Vikhram S

Contact Information

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Education

Saveetha Engineering College (SEC) ; Chennai, India — 2022 – 2026
Bachelor of Engineering in Electronics and Communication Engineering — CGPA : 8.10/10
Courses : Digital Image Processing, Artificial Intelligence, Natural Language Processing, Random Processes and Statistics, Linear Algebra and Numerical Methods, Big Data Mining and Analysis

Thesis on "Explainable Vision–Language Generative Framework for Radiology Report Synthesis and Interpretation"

Experience

Independent Research — India
Survival Modeling with RNA-seq and Clinical Data under Site-Held-Out Validation (TCGA-KIRC) — *2026*

- Designed and carried out a protocol-driven test of whether the added prognostic value of bulk RNA-seq over age, sex, and stage in clear cell renal cell carcinoma survives withholding entire tissue source sites; the protocol was version-controlled and frozen before model fitting, and every deviation, including one post hoc model correction, is logged and archived.
- Found that an elastic-net Cox model with RNA-seq added a concordance gain over a clinical Cox baseline ($N = 529$, 173 deaths, 20 sites) of $+0.024$ under random and $+0.031$ under site-held-out cross-validation; the gain was not attenuated, though the study could not distinguish the two schemes ($\Delta\Delta C = -0.007$, 95% CI -0.044 to $+0.018$).
- Quantified the optimism from whole-cohort gene selection ($+0.011$ in C-index), added negative controls, leave-one-group-out validation, and a sensitivity audit, and stated the limits (single cohort, linear models, no histologic grade in the baseline); released code, a claims registry, and a TRIPOD+AI checklist on GitHub.

Saveetha Engineering College (SEC), Chennai — India
Undergraduate Thesis (Advisor : Dr. Jeffin Gracewell J) — *Aug 2025 – Feb 2026*

- Proposed an explainable vision–language framework for chest X-ray report generation and built a ViT–BioGPT prototype on MIMIC-CXR; later found that the prototype does not condition on the image (a silent cross-attention omission in the BioGPT decoder), documented it with a reproducible check, and published it as a labelled negative result.
- Corrected pipeline with cross-attention, held-out evaluation, and a shuffled-image control planned.

Virtusa Consulting Services Pvt Ltd — India
Intern – Delivery (AI / Machine Learning) — *Mar 2026 – May 2026*

- Led a four-member team, selected through Virtusa's **Jatayu Hackathon 2026** (Agentic AI theme), to build **CodePerfect Auditor**, a healthcare revenue-integrity auditing system that reconciles clinical documentation against billed ICD-10 and CPT codes and links every flagged discrepancy to the exact supporting sentence in the source clinical note.
- Built the clinical NLP extraction layer, configurable code-mapping logic, and discrepancy-detection engine; shipped as a **FastAPI** service with a **Streamlit** reviewer interface.

Publications

Jeffin Gracewell J, Yuvaraj J, **Vikhram S***. **Explainable SLM-Guided Vision–Language Model for Multi-Class Skin Lesion Recognition.** In Proc. ICICC '26, Lecture Notes in Networks and Systems, vol. 2045, Springer, Cham, 2026. [Paper] [Code]

Vikhram S. **Does RNA-seq Add Prognostic Value Beyond Clinical Variables Under Site-Held-Out Validation in TCGA-KIRC?** Research Square preprint, 2026. [Paper] [Code]

Vikhram S*, Jeffin Gracewell J. **NariRaksha: Gender-Responsive AI for Women's Safety** (case study). Case Study 12, pp. 68–70, in Real-World Impact of AI and Gender Empowerment: Casebook on AI and Gender Empowerment, India AI Impact Summit 2026 Compendium, IndiaAI Mission (MeitY, Government of India) & UN Women, 2026. [Paper]

Vikhram S. **IndianConstitution: Building Trustworthy, Reproducible, and Type-Safe Infrastructure for Legal NLP and Constitutional Informatics.** Zenodo preprint, 2026. [Paper] [Code]

Vikhram S. **Beyond Code Smell: Measuring Architectural Conformance Drift in Long-Horizon Autonomous Coding Agent Sessions – A Pilot Study.** Zenodo preprint, 2026. [Paper] [Code]

*Corresponding author.

Academic Service

Invited Reviewer, **SocialAgent @ NeurIPS 2026** (Second Workshop on Large Language Models for Social Reasoning and Simulation) Sept 2026

Peer Review — OpenReview; reviewed submissions on LLM-based social reasoning and simulation during the workshop's review cycle (Sept 7–23, 2026)

Projects

Auditing a ViT–BioGPT Report Generator, Hugging Face 2025

- Diagnosed a silent cross-attention omission (decoder ignores the image) via parameter, checkpoint-size, and gradient checks; published a reproducible test and a corrected-pipeline plan with shuffled-image control.

Gender-Responsive Safety Model for Women's Safety (NariRaksha-3B), Hugging Face 2026

- Released a QLoRA fine-tune of **Qwen2.5-3B-Instruct** on NariRaksha-100K as an open research artifact for structured women's-safety assessments; the model card documents the absence of a held-out evaluation and the likely memorization.

Benchmark Dataset for Indic Constitutional and Legal Reasoning, Hugging Face 2025

- Constructed a benchmark dataset for evaluating constitutional and legal reasoning in Indic-language NLP systems.

Open-Source Constitutional Intelligence Platform for India, GitHub / PyPI 2025

- Built an open source platform for semantic retrieval and structured analysis of India's constitutional texts; over **38,000 cumulative downloads** on PyPI.

Teaching

Invited Lectures, five departmental lectures, Saveetha Engineering College Mar – Oct 2025

- AI Agents and Open Source Ecosystems for Intelligent Software Development, Mar 2025
- Quantum Machine Learning: Emerging Directions in Hybrid Quantum–Classical AI, May 2025
- Financial Machine Learning: Macroeconomic Indicators, Forecasting, and Predictive Analytics, May 2025
- Software Engineering Practices for Reproducible Deep Learning Systems, Sept 2025
- Cyber Law and Compliance Series: Security, Contracts, and IP in the Digital Era, Oct 2025

Honors and Awards

- **Selected Contributor, India AI Impact Summit 2026 Casebook on AI and Gender Empowerment** (IndiaAI Mission, MeitY and UN Women): first author of *NariRaksha: Gender-Responsive AI for Women's Safety* (Case Study 12), one of **23 case studies selected from over 235 submissions by innovators in more than 50 countries** after multi-stage double-blind review; presented with a copy of the casebook at the official publication launch by Sudeshna Mukherjee, Head of Communications, UN Women India Country Office. 2026
- **First Prize, Department of ECE Project Contest 2026**, Saveetha Engineering College: awarded for *Explainable Vision–Language Generative Framework for Radiology Report Synthesis and Interpretation*; certified by the Head of Department, Department of Electronics and Communication Engineering. 2026

Positions of Responsibility

Advisor, Machine Learning Tech Society, Saveetha Engineering College Aug 2025 – Present

- Invited by the Dean of Information and Communication Technology to provide academic and technical leadership for the institution's machine learning community.
- Led programs on **AI Agents**, **Quantum Machine Learning**, and **Financial Machine Learning**, and mentored student researchers.

References

Dr. Jeffin Gracewell J, Saveetha Engineering College, Chennai jeffingracewellj@saveetha.ac.in

Associate Professor, Department of ECE

Dr. Srigitha S Nath, Saveetha Engineering College, Chennai srigithasnath@saveetha.ac.in

Associate Dean (ECE, BME, and Medical Electronics); Professor and Head, Department of ECE